

Project Artemis: AI Document Q&A; Technical Specification

Section 1: Architecture Overview

Project Artemis is a Retrieval-Augmented Generation (RAG) system designed to process PDF documents and answer user queries with precise citations. The system utilizes Chroma DB as the vector store, sentence-transformers for local embedding generation, and supports both local Ollama (Gemma model) and cloud Gemini API as LLM backends.

Section 2: Data Ingestion Pipeline

Documents are loaded using PyPDFLoader, split into overlapping chunks of 1000 characters with a 200-character overlap using RecursiveCharacterTextSplitter. Chunks are embedded into 384-dimensional vectors and indexed in Chroma with metadata tracking source filename and page number.

Section 3: Evaluation & Citations

Every response generated by Project Artemis includes explicit citations indicating the source document name and page number. Retrieved context chunks are displayed in the user interface to ensure hallucination reduction and answer verifiability.